

Heavy-tailed neuronal connectivity arises from Hebbian self-organization

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The connections in networks of neurons are heavy-tailed, with a small number of neurons connected much more strongly than the vast majority of pairs. However, it remains unclear whether this heavy-tailed connectivity emerges from simple underlying mechanisms. Here we propose a minimal model of synaptic self-organization: connections are pruned at random, and the synaptic strength rearranges under a mixture of preferential and random dynamics. Under these generic rules, networks evolve to produce distributions of connectivity strength that are asymptotically scale-free, with a power-law exponent that depends only on the probability of preferential (rather than random) growth. Extending our model to include neuronal activity and Hebbian plasticity, we find that clustering in the network also emerges naturally. We confirm these predictions in the connectomes of several animals, suggesting that heavy-tailed and clustered connectivity may arise from general principles of network self-organization rather than mechanisms specific to individual species or systems.

Neurons communicate and interact via an intricate web of synaptic connections. Although this wiring is constantly shifting and rearranging^{1–3}, its structure is far from random, with synaptic connectivity concentrated among a select few pairs of neurons⁴. Indeed, the distribution of connection strengths is heavy-tailed^{5–9}, with a sparse network of strong connections dominating in a sea of weak interactions. These strong connections form the backbone of neuronal circuitry that is responsible for unparalleled feats of information processing¹⁰, from learning^{1,11–13} and memory^{14–17} to sensing^{18–22}, communication^{23,24} and motor control^{25–28}. Yet, despite the importance of strong connections in guiding neural function, there remain fundamental questions about how heavy-tailed connectivity arises in neuronal populations. Does it require complicated biophysical processes specific to each species or neural system? Or, instead, can heavy-tailed connectivity emerge generically from principles of network self-organization?

Heavy-tailed neuronal connectivity

To answer these questions, we begin by comparing the distributions of connection strengths across several species. Gathering sufficient data

to characterize the tails of connection strength distributions presents a major challenge, requiring cellular-resolution maps of large volumes of nervous tissue²⁹. However, recent advances in high-throughput electron microscopy and image processing have allowed unprecedented snapshots of the physical wiring connecting thousands to tens of thousands of neurons^{5,11,12,18,19,25,26}. Given this experimental progress, it is now possible to investigate the general mechanisms underlying large-scale neuronal connectivity.

Consider, for example, the partial connectome of the *Drosophila* central brain (Fig. 1a), with 14 million synapses linking over 21 thousand neurons (approximately one quarter of the fly's brain)⁵. Despite the large number of synapses, 99% of neuron pairs remain unconnected; and even among connected neurons, 50% are only linked by a single synapse (Fig. 1b). Meanwhile, some pairs of neurons are connected by over one thousand individual synapses; these rare but extremely strong connections comprise the heavy tail of synaptic connectivity (Fig. 1b). By contrast, if we randomize the placement of synapses in the network, then connectivity strength drops off super-exponentially as a Poisson distribution (Fig. 1b), and the backbone of strong connections

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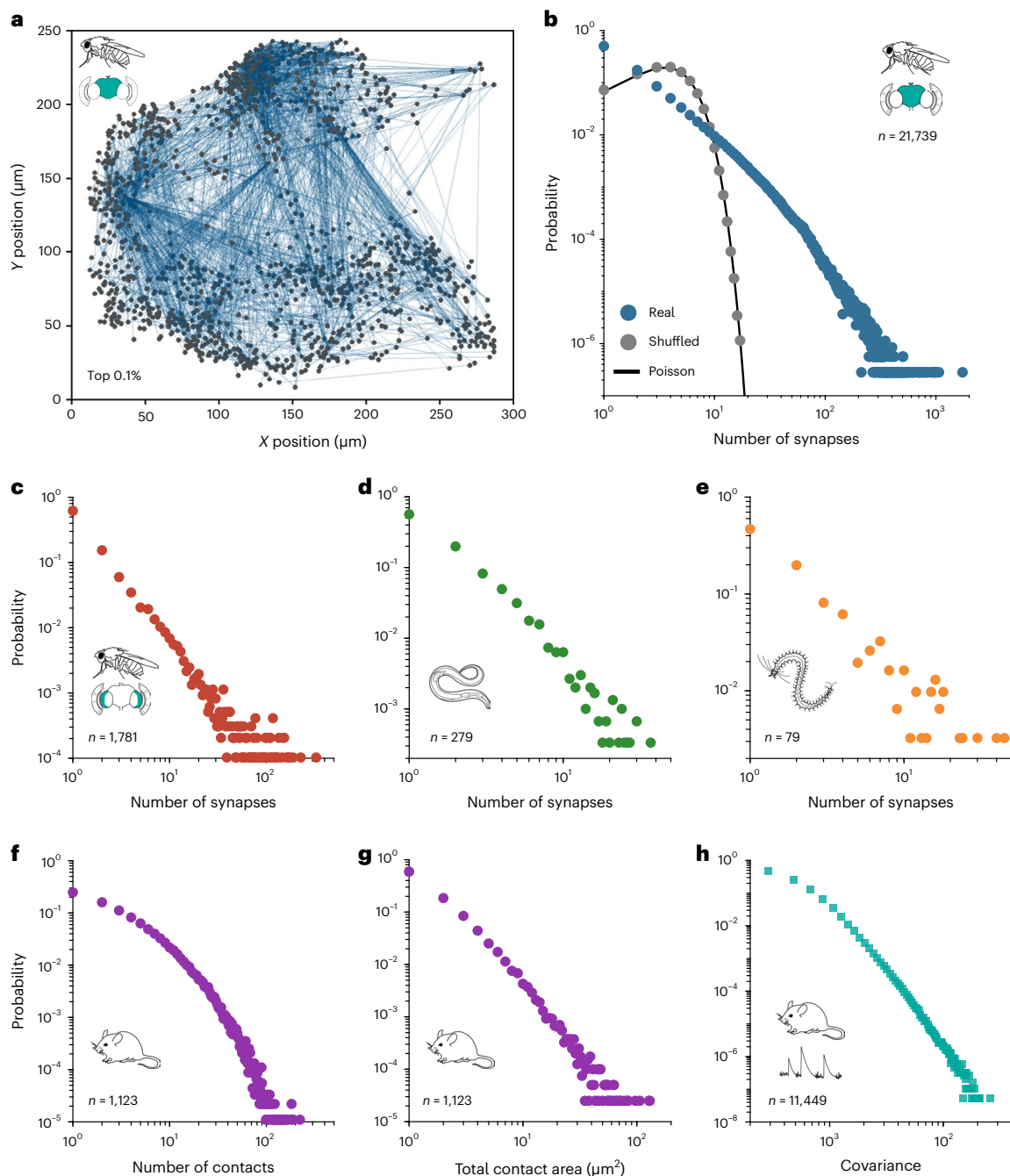


Fig. 1 | Heavy-tailed connectivity in neuronal connectomes. **a**, Network of the strongest 0.1% of connections in the connectome of the *Drosophila* central brain, traced using volume electron microscopy (Methods)⁵. **b**, Distribution of the number of synapses in each connection of the *Drosophila* central brain (blue) and after shuffling the synapses among the neurons (grey). The solid line indicates a Poisson distribution. **c–e**, Distributions of the number of synapses in each connection of the *Drosophila* optic medulla (**c**)¹⁹, the entire connectome of the roundworm *C. elegans* (**d**)^{25,30} and the sensory motor circuit of the annelid

Platynereis (**e**)²⁶. **f, g**, Distributions of the number of contacts (**f**) and total contact area (rounded to the nearest μm^2) (**g**) between neurons in the inner plexiform layer of the mouse retina¹⁸. **h**, Distribution of covariances in neuronal activity in the mouse visual cortex while responding to natural images, recorded using two-photon calcium imaging (Methods)³². Square data points are used to differentiate from the structural connectivity in panels **a–g**. See Table 1 for the details of each network.

vanishes. We observe similarly heavy-tailed connectivity in the *Drosophila* optic medulla (Fig. 1c)¹⁹, the connectome of the roundworm *Caenorhabditis elegans* (Fig. 1d)^{25,30} and the sensory motor circuit of the annelid *Platynereis* (Fig. 1e)²⁶.

Thus far we have focused on the connectomes of invertebrates, in which stronger connections are formed by increasing the number of synapses in parallel⁵. In vertebrates, meanwhile, connections are

strengthened by increasing both the number and size of individual synapses^{5,31}. In the mouse retina, for example, we find that the distribution of the number of physical contacts between neurons (a proxy for synaptic connections) is heavy-tailed (Fig. 1f), but not to the same degree as invertebrate connectivity (Fig. 1b–e)¹⁸. This accounting, however, fails to include information about the sizes of individual contacts or synapses, which themselves have been observed to follow heavy-tailed

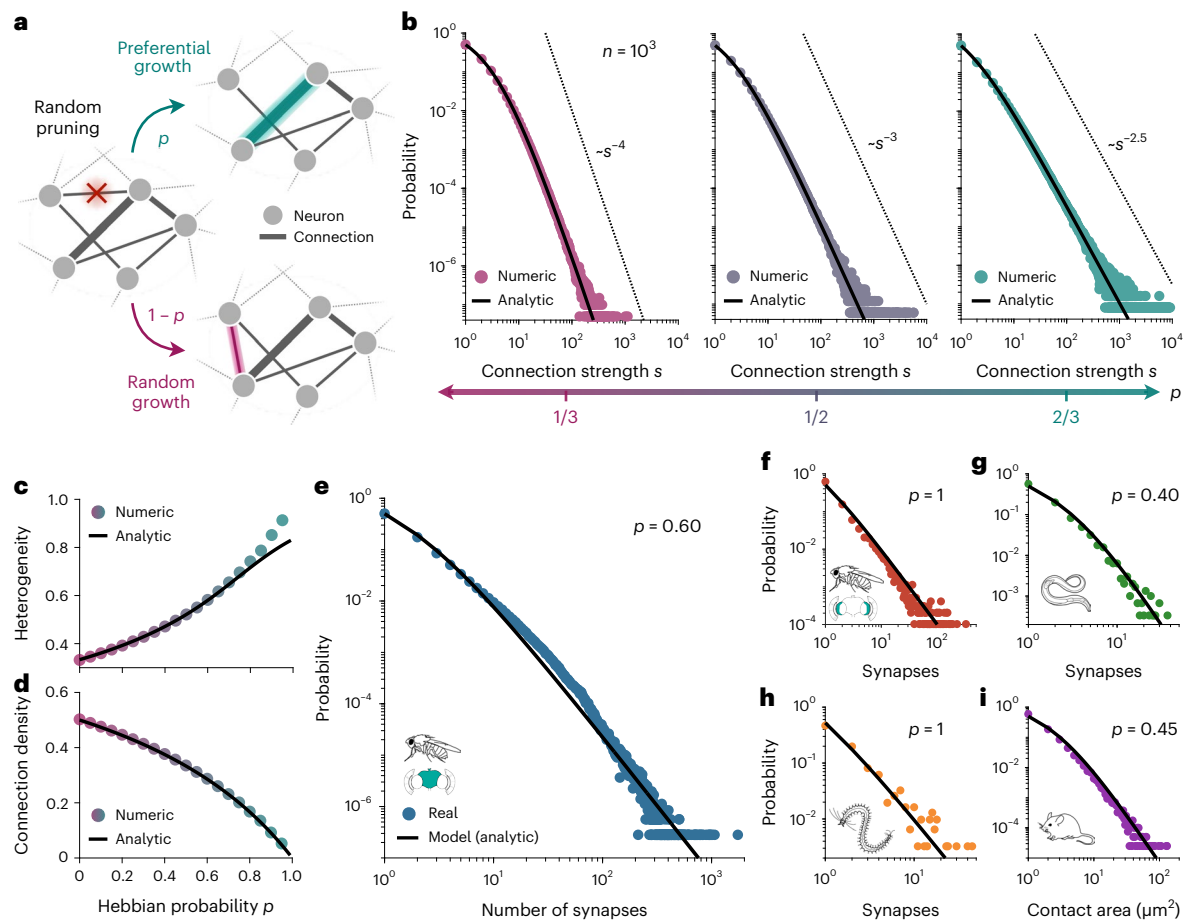


Fig. 2 | Preferential growth produces scale-free connection strengths.

a, Minimal model of preferential network dynamics. Left: at each time step, a random connection is pruned and top: each unit of pruned strength is redistributed either preferentially (with probability p) or bottom: randomly (with probability $1-p$). **b**, Distributions of connection strengths for increasing probabilities of preferential growth p in networks of $n = 10^3$ neurons and average connection strength $\bar{s} = 1$. Data points depict numeric simulations (Methods), solid lines represent the analytic prediction in equation (1) and dashed lines

reflect power laws with the predicted exponent $\gamma = 1 + \frac{1}{p}$. **c–d**, Connection heterogeneity (**c**) and density (**d**) as functions of p , computed using numeric simulations (data points) and the analytic distribution (lines). Connection density is the fraction of all $n(n-1)$ possible connections with non-zero strength. **e–i**, Analytic model predictions from equation (1) with $\bar{s} = 1$ (to match the simulations in panels **b–d**) and p fit to the connectivity distributions (Methods) of the *Drosophila* central brain (**e**), *Drosophila* optic medulla (**f**)¹⁹, *C. elegans* connectome (**g**)^{25,30}, *Platynereis* sensory motor circuit (**h**)²⁶ and mouse retina (**i**)¹⁸.

(often log-normal) distributions^{4,9,11}. Indeed, if we consider the total strengths of connections—that is, the total contact area between neurons—we find a distribution that once again resembles those observed in invertebrates (Fig. 1g). Finally, leveraging recent large-scale recordings of neuronal activity, we can examine the strengths of ‘functional’ connections (or correlations) between neurons. In a large-scale recording of the mouse visual cortex responding to natural images³², we find that the distribution of covariances in activity reflects the heavy-tailed structural connectivity between neurons (Fig. 1h), a result that we confirm in different mice and across distinct stimuli (Supplementary Fig. 1).

Model of network self-organization

The above results suggest that heavy-tailed connectivity is a general feature of neuronal networks. It is tempting, then, to seek an equally general explanation. In the study of complex systems, heavy-tailed distributions often arise from preferential attachment or rich-get-richer mechanisms. However, whereas existing network models focus on distributions of degrees (or node strengths)^{33,34}, here we study connection strengths. In the context of synaptic connectivity, a form of preferential attachment already lies at the core of network formation: Hebbian plasticity, wherein strongly connected neurons are more likely

to co-fire, leading to even further growth^{2,31,35,36}. But can the preferential attachment of connectivity on its own (that is, even before including activity dependence and other biophysical effects) explain the heavy tails observed in real connectomes?

To answer this question, here we present a minimal model of network formation in which connectivity self-organizes under a mixture of preferential and random dynamics. Specifically, we begin with n neurons connected by S units of connection strength (for example, each unit could represent one synapse, $1 \mu\text{m}^2$ of contact area or even non-integer measures of strength). The network is defined by a connectivity matrix A , where A_{ij} is the directed connection strength from neuron i to neuron j , such that $\sum_j A_{ij} = S$. At each point in time, one connection is selected at random to be pruned, resulting in $A_{ij} \rightarrow 0$ (Fig. 2a, left). Each unit of pruned strength is then independently redistributed to a new pair of neurons (such that $A_{ij} \rightarrow A_{ij} + 1$) in one of two ways: (1) with probability p , a connection is preferentially selected for growth with probability A_{ij}/S proportional to its current strength (Fig. 2a, top); or (2) with probability $1-p$, a random pair of neurons is selected for growth (Fig. 2a, bottom). In this way, the total connection strength S remains constant, with the wiring between neurons simply rearranging over time. Moreover, we note that after specifying the number of neurons

n and the total connection strength S , the model only contains a single parameter p , representing the probability of preferential rather than random growth.

Ultimately, we wish to understand how the above dynamics impact the distribution $P(s)$ of connection strengths s . Using the master equation method (which has previously been applied to study degree distributions³³), one can show that as the network evolves, the connection strengths approach an analytic steady-state distribution

$$P(s) \propto \frac{\Gamma\left[s + \bar{s}\left(\frac{1}{p} - 1\right)\right]}{\Gamma\left[s + \bar{s}\left(\frac{1}{p} - 1\right) + 1 + \frac{1}{p}\right]}, \quad (1)$$

where $\bar{s} = \frac{S}{n(n-1)}$ is the average connection strength and $\Gamma(\cdot)$ is Euler's gamma function (Methods). Throughout the paper, we normalize all distributions to run over positive connection strengths $s > 0$ (that is, excluding null connections with $s = 0$). In the limit of strong connections $s \gg \bar{s}/p$, the above distribution falls off as a power law $P(s) \approx s^{-\gamma}$, with a scale-free exponent $\gamma = 1 + \frac{1}{p}$ that depends only on the probability of preferential growth p (Methods). We confirm the analytic distribution in equation (1) and the power-law tail in numeric simulations (Fig. 2b).

As the proportion of preferential growth p decreases, the network becomes more random, limiting the heavy tail in $P(s)$ and increasing the exponent γ (Fig. 2b, left). By contrast, if preferential growth dominates, strong connections are more likely to become even stronger, yielding a heavier tail and lower exponent γ (Fig. 2b, right). To quantify the heavy-tailedness of power-law-like distributions, a common measure is the heterogeneity $\frac{1}{2} \langle |s - s'| \rangle / \langle s \rangle$, where $\langle \cdot \rangle$ represents an average over $P(s)$, and $\langle |s - s'| \rangle$ is the average absolute difference in connection strengths^{37–39}. While the variance of a power-law distribution diverges for $\gamma \leq 3$ (or $p \geq 0.5$ in our model), the heterogeneity (which is normalized to lie between zero and one) is well-defined for all $\gamma > 2$ (that is, all values of $p < 1$). As p increases, we see that connection strengths become more heterogeneous (Fig. 2c) while the density of connections decreases (Fig. 2d). Thus, preferential dynamics lead to the formation of sparse networks with heavy-tailed connection strengths—precisely the features observed in real connectomes^{5,11,12,18,19,25,26,29}.

We can even fit the preferential probability p (the only free parameter in the model) to the connectivity distributions of individual connectomes; for example, the *Drosophila* central brain is best described as arising from a combination of 60% preferential growth and 40% random growth (Fig. 2e). In fact, despite only fitting one parameter, our simple model provides a surprisingly good description of all the networks considered (Fig. 2e–i), with the proportion of preferential growth ranging from 40% for *C. elegans* (Fig. 2g) to 100% for the *Drosophila* optic medulla (Fig. 2f) and *Platynereis* sensory motor circuit (Fig. 2h). The success of our model is even more surprising when considering that each network has distinct genetic constraints on network structure. Together, these results demonstrate that model neurons can self-organize under a minimal rich-get-richer mechanism to produce the kinds of sparse, heavy-tailed networks observed in real connectomes.

Introducing neuronal activity

The model above generates networks with realistic connection strengths, even without including the effects of neuronal activity. In real neuronal networks, however, one expects synaptic connections to evolve not just on their own, but also in response to correlations in activity—a process known as Hebbian plasticity^{2,31,35,36}. By including this feedback between network structure and function, here we will show that additional properties of neuronal connectomes emerge naturally. To introduce correlations between neurons, we must begin with a model of neuronal activity. While one can implement varying degrees

of biological realism^{10,40}, for simplicity here we consider artificial neurons that, between each network update, reach steady-state activities x_i defined by the self-consistent equations $x_i = \tanh(\beta \sum_j A_{ij} x_j)$, where $\beta \geq 0$ parameterizes the strength of interactions, and the connections A_{ij} are implicitly normalized by $N\bar{s}$ (Methods). This model is a mean-field approximation to the Ising model^{41–43}, which has been shown to capture key features of neuronal populations^{44,45}. Moreover, the activities x_i can be mapped directly to firing rates (Methods), making the model equivalent to an artificial recurrent neural network^{10,43,46}. The primary advantage of this model is the fact that, if the connections A_{ij} are symmetric (as will be assumed in what follows), then we can use the fluctuation–dissipation theorem to derive a succinct analytic form for the covariances (Methods): $C = \beta(I - \beta DA)^{-1}D$ where $D_{ij} = (1 - x_i^2)\delta_{ij}$ is a diagonal matrix and I is the identity.

We are now prepared to extend our network dynamics to include activity-dependent plasticity. The random pruning (Fig. 2a, left) and random growth (Fig. 2a, bottom) remain unchanged, but we replace preferential growth with Hebbian growth: rather than selecting a connection with probability proportional to its current strength A_{ij} (Fig. 2a, top), we instead select a pair of neurons with probability proportional to their covariance C_{ij} (Methods). To gain intuition for this activity-dependent model, we note that $x_i = 0$ in the limit of weak interactions $\beta \ll 1$, yielding covariances

$$C = \beta(\beta A + \beta^2 A^2 + \beta^3 A^3 + \dots) = \beta \sum_{n=1}^{\infty} (\beta A)^n \quad \text{for } i \neq j. \quad (2)$$

To lowest order in β , the covariances are proportional to the connection strengths $C \propto A$, and therefore Hebbian plasticity reinforces existing connections; that is, Hebbian growth acts exactly as preferential growth in the original model (Fig. 3a, top). Thus, the original activity-independent dynamics (Fig. 2) are precisely the weak-interaction limit of the more general activity-dependent model.

To second-lowest order in β , correlations are proportional to A^2 , and so Hebbian plasticity connects neurons that share common neighbours in the network (that is, neurons that interact via paths of length two). In this case, activity-dependent growth leads to the formation of triangles (Fig. 3a, middle), which, as we will see, produces clustering in the topology of connections. In general, as the interaction strength β increases towards one (the critical interaction strength for random networks; Supplementary Information section 3), Hebbian plasticity reinforces interactions of increasing topological distance, thereby producing longer and longer loops in the network (Fig. 3a, bottom). Importantly, we expect these results (and those below) to qualitatively generalize to other models of neuronal activity. As long as the correlations between neurons decrease with topological distance in the network—that is, for any model of neuronal activity where direct connections produce the strongest correlations, length-two connections produce weaker correlations, and so on—our model of Hebbian plasticity will still reinforce existing connections (to first order) followed by progressively longer loops in the network (Fig. 3a).

Clustering arises from activity-dependent plasticity

Simulating the above dynamics, we can examine how Hebbian plasticity impacts both the strength and topology of connections. In the limit of weak interactions, because Hebbian plasticity becomes equivalent to preferential growth, the connection strengths converge to the heavy-tailed distribution in equation (1) (Fig. 3b). Moreover, since $C \propto A$, the model also produces heavy-tailed correlations, just as observed in large-scale recordings of neuronal activity (Fig. 1h and Supplementary Fig. 2). By contrast, as β increases, Hebbian growth fails to reinforce existing connections (Fig. 3a), thereby constraining the heavy tail of connection strengths (Fig. 3b). Indeed, we find that the heterogeneity of connection strengths increases with the Hebbian probability p and

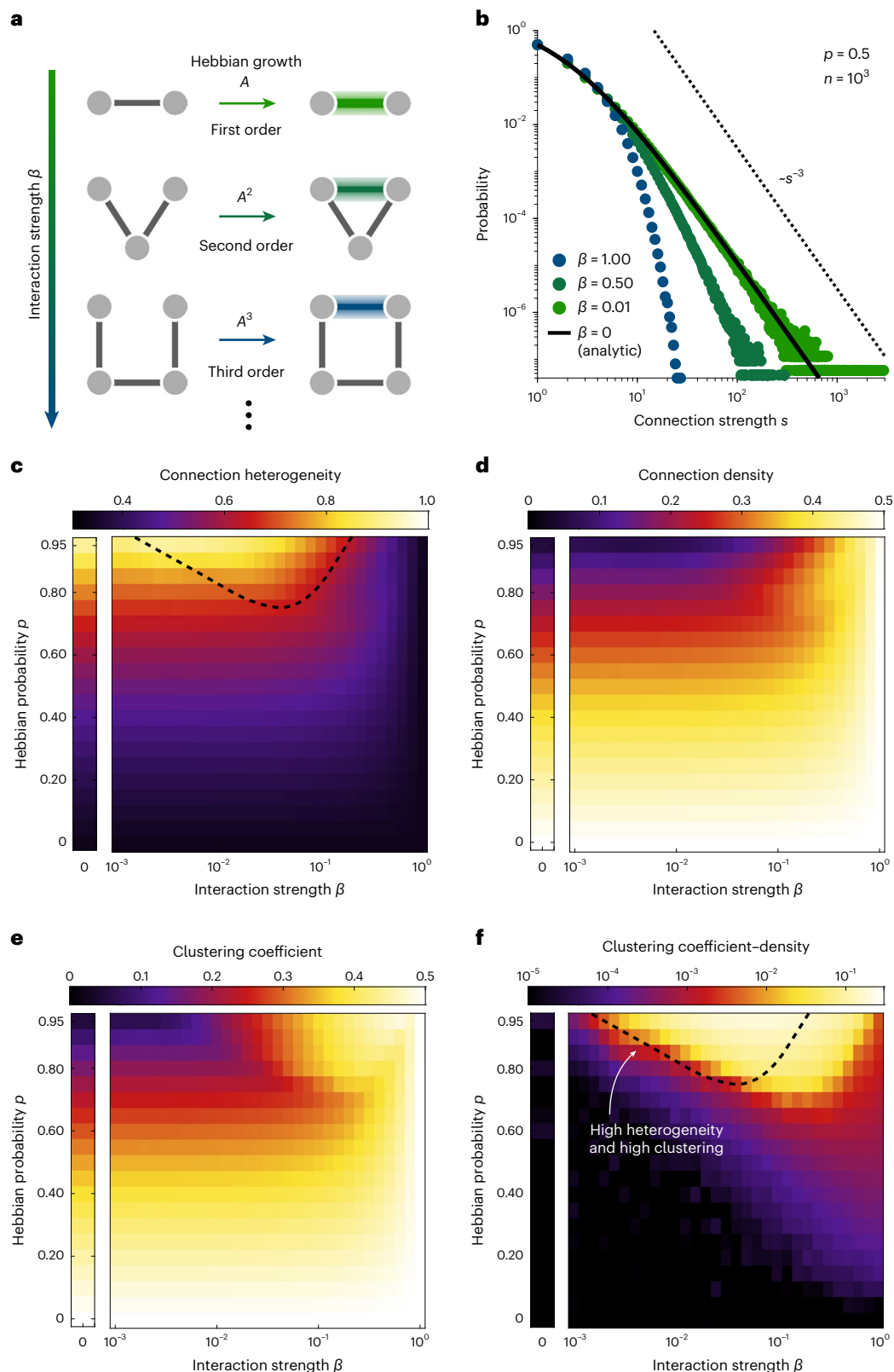


Fig. 3 | Activity-dependent dynamics give rise to network clustering.

a, Expanding Hebbian growth in the limit of weak interactions. Top: to lowest order in the interaction strength β , Hebbian growth reinforces existing connections; middle: to second order, Hebbian growth reinforces interactions of length two, forming triangles; bottom: in general, to l th order, Hebbian growth reinforces interactions of length l , thus closing loops of length $l + 1$. **b**, Distributions of connection strengths for different interaction strengths β in networks of $n = 10^3$ neurons, Hebbian probability $p = 0.5$, and average connection strength $\bar{s} = 1$. Solid line depicts the analytic activity-independent prediction in equation (1) and

dashed line illustrates a power law with exponent $\gamma = 1 + \frac{1}{p} = 3$. **c–f**, Phase diagrams of connection heterogeneity (**c**), connection density (**d**), clustering coefficient (**e**) and the difference between clustering coefficient and connection density (**f**) as functions of the Hebbian probability p and interaction strength β for simulated networks of $n = 10^3$ neurons and average connection strength $\bar{s} = 1$ (Methods). Left: in each panel, for $\beta = 0$, we plot values for the original activity-independent model (Fig. 2). In panels **c** and **f**, dashed lines indicate the region of parameter space with simultaneously heavy-tailed connection strengths and highly clustered topologies.

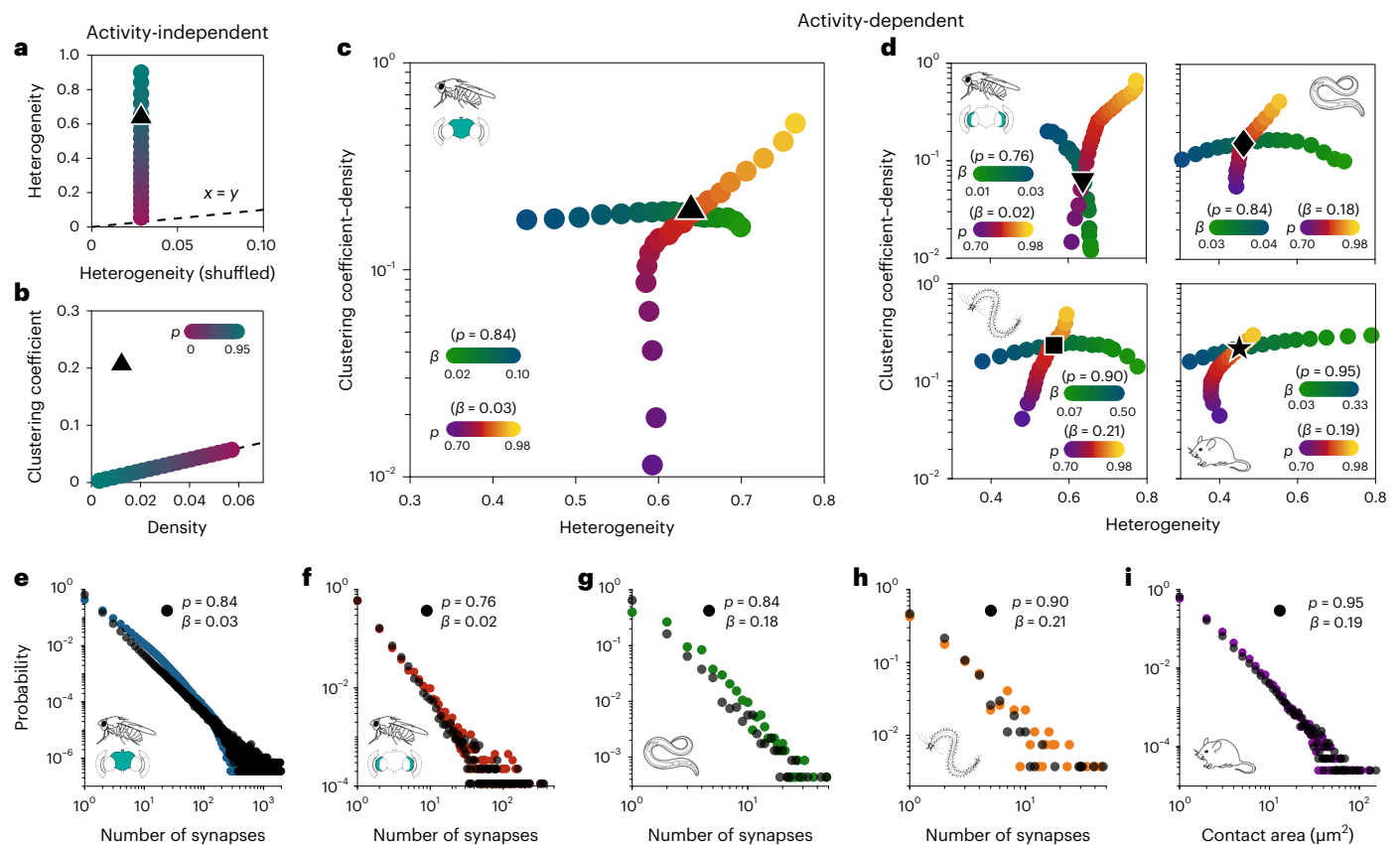


Fig. 4 | Capturing heavy-tailed and clustered connectivity in real connectomes. **a**, Heterogeneity of connection strengths in the *Drosophila* central brain before and after shuffling the synapses among the neurons. Coloured points depict the activity-independent model while sweeping over p for networks of $n = 10^3$ neurons and average connection strength $\bar{s}$ matching the *Drosophila* central brain (Methods). **b**, Clustering coefficient versus connection density for the same networks as **a**. **c**, Clustering coefficient minus connection density (that is, the clustering beyond that of a network with randomized connections) versus connection heterogeneity. Coloured points illustrate the activity-dependent model (with $n = 10^3$ and $\bar{s}$ matching the *Drosophila* central brain) while sweeping over the interaction strength β (blue/green) or Hebbian probability p (purple/yellow). For $p = 0.84$ and $\beta = 0.033$, the activity-dependent

dynamics yield the same connection heterogeneity and clustering coefficient (minus density) as the *Drosophila* central brain. To improve sampling, data points are averaged across adjacent parameter values. **d**, Top left: same as **c** but for the *Drosophila* optic medulla¹⁹, top right: *C. elegans* connectome^{25,30}, bottom left: *Platynereis* sensory motor circuit²⁶ and bottom right: mouse retina¹⁸. **e–i**, Distributions of connection strengths in the activity-dependent model with parameters p and β chosen to match the heterogeneity and clustering coefficient (minus connection density) of the *Drosophila* central brain (**e**)⁵, *Drosophila* optic medulla (**f**)¹⁹, *C. elegans* connectome (**g**)^{25,30}, *Platynereis* sensory motor circuit (**h**)²⁶ and mouse retina (**i**)¹⁸. Across all panels, to compute clustering and compare against model networks, the real connectomes are symmetrized, such that the connectivity is defined by $A + A^T$ (Methods).

decreases with the interaction strength β (Fig. 3c). Conversely, the density of connections decreases with p and increases with β (Fig. 3d), such that sparse, heavy-tailed connectivity arises for large Hebbian probabilities and weak to intermediate interactions.

While to first order in β , Hebbian plasticity yields heavy-tailed connectivity, as discussed above, the second-order effect is the formation of triangles (Fig. 3a). Notably, real neuronal networks are known to exhibit a high density of triangles, a property known as clustering^{4,47–49}. Clustering is typically quantified using the ratio n_{Δ}/n_{λ} of the number of neuron triangles n_{Δ} to the number of triplets n_{λ} (sets of three neurons with at least two connections). For random networks, this clustering coefficient is identical to the density of connections; and in fact, we see that the clustering coefficient (Fig. 3e) closely tracks the connection density (Fig. 3d) across most values of p and β . In particular, as interactions vanish ($\beta \rightarrow 0$), the original activity-independent dynamics (Fig. 2) yield random network topologies (for all values of p), such that the clustering coefficient becomes identical to the connection density (Fig. 3f, left). Thus, if interactions are too strong, then connection strengths lose their heavy tail (Fig. 3b,c), and if interactions are too weak, then the network fails to develop clustering (Fig. 3f). Yet for intermediate interaction strengths β , the activity-dependent

dynamics can produce both heavy-tailed connection strengths and highly clustered topologies (Fig. 3c,f, dashed lines), two fundamental features of real neuronal networks^{4–9,47–49}.

The activity-dependent dynamics generate realistic network features, but can they quantitatively reproduce the properties of specific neuronal connectomes? For example, in the *Drosophila* central brain (Fig. 1a,b), we find that the connections are more heterogeneous (Fig. 4a) and more clustered (Fig. 4b) than comparable random networks; and we confirm the same results across all of the connectomes analysed in Fig. 1 (Supplementary Fig. 3). Simulating networks with the same average connection strength $\bar{s}$ as the *Drosophila* central brain, the original activity-independent model (Fig. 2) can capture the connection heterogeneity observed in the data (Fig. 4a), but (due to its random topology) fails to produce realistic clustering (Fig. 4b). However, by generalizing to activity-dependent dynamics, the networks can simultaneously attain both the connection heterogeneity and clustering observed in the *Drosophila* connectome (Fig. 4c). In fact, by fitting only two parameters (the Hebbian probability p and interaction strength β), the activity-dependent dynamics quantitatively reproduce the connection heterogeneity and clustering in all of the connectomes analysed (Fig. 4d). Moreover, for the same values of p and β , we find

that the activity-dependent model accurately predicts the distributions of connection strengths in real networks (Fig. 4e–i). Thus, by extending the network dynamics to include Hebbian plasticity, the wiring between neurons self-organizes to produce not only the heavy-tailed connection strengths but also the clustering observed in real networks.

Discussion

Across a diverse range of species, the strengths of neuronal connections are similarly heavy-tailed (Fig. 1), suggesting a common underlying mechanism. However, generating realistic features of neuronal connectivity from simple rules is notoriously difficult^{50,51}. Here we develop a minimal model in which synaptic connectivity self-organizes under a mixture of preferential and random dynamics. The resulting networks exhibit distributions of connectivity strength that are asymptotically scale-free, with an exponent $\gamma = 1 + \frac{1}{p}$ that depends only on the proportion p of preferential growth (Fig. 2). Furthermore, by generalizing the model to include Hebbian (activity-dependent) plasticity, the network dynamics naturally give rise to clustering (Fig. 3), another prominent feature of neuronal connectivity (Fig. 4).

Given the simplicity of the model, future work can immediately begin to include more realistic plasticity mechanisms, which, in turn, may produce additional properties observed in neuronal connectomes. For example, rather than updating the synaptic connectivity based on equal-time correlations, one could further generalize the activity-dependent model to include spike-time-dependent plasticity, wherein synapses are updated based on time-delayed correlations^{3,35}. These temporal asymmetries will naturally generate asymmetries in the connections themselves, a central property of real connectomes^{4–9,11,12,18,19,25,26,47}. Moreover, for simplicity, our network dynamics (1) include global information about the total connectivity strength S , (2) assume a uniform probability of pruning any given connection (Fig. 2a, left) and (3) allow all possible connections between neurons. Future work can introduce additional biological realism by restricting to local information about network topology, considering non-uniform pruning and constraining the allowed connections—three features of real neuronal plasticity^{52,53}.

In addition to new plasticity mechanisms, one can also directly implement more biophysically realistic models of neuronal activity in the activity-dependent model⁴⁰. Here, to allow for analytic derivations, we have focused on a statistical physics model of recurrent nonlinear dynamics. Ising models such as these are capable of reproducing the firing rates and correlations observed in real neuronal populations^{44,45}. These maximum entropy models have mostly been applied to static network topologies, opening the door for investigations of dynamic connectivity. Alternatively, in future work one could study more biologically inspired models of spiking activity or neural oscillations^{10,54}. Importantly, for any model where the correlations between neurons decrease with distance in the network, one should still expect our framework to produce networks with both heavy-tailed distributions of connection strength and high clustering (Fig. 3). Together, these results demonstrate how two fundamental features of connectomes—heavy-tailed connectivity and clustering—can arise from simple network dynamics, providing a framework for future investigations into the self-organization of neuronal connectivity.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41567-023-02332-9>.

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Methods

Neuronal connectivity data

The datasets analysed in this paper were each previously published and are openly accessible^{5,18,19,25,26,32}. All connectomes were sectioned and imaged at the neuron level using electron microscopy^{5,18,19,25,26}. The neurons and synapses (or contacts for the mouse retina) were then traced using a mixture of manual labelling and automated image processing. For further experimental details concerning the *Drosophila* central brain (Fig. 1a,b)⁵, *Drosophila* optic medulla (Fig. 1c)¹⁹, *C. elegans* (Fig. 1d)^{25,30}, *Platynereis* sensory motor circuit (Fig. 1e)²⁶ and mouse retina (Fig. 1f,g)¹⁸, see the respective references. We have made the lists of neurons and connections between them openly accessible in a common format ('Data Availability').

When comparing against the activity-independent model (Figs. 1 and 2), the connections A_{ij} are directed, reflecting the number of synapses (or for the mouse retina, the area of contacts) from one neuron i to another j . For the mouse retina, we round the total area of each connection to the nearest μm^2 to create discrete units of connection strength. When comparing against the activity-dependent model (Fig. 4), we use symmetrized versions of the connectomes, where the connections $A + A^T$ reflect the number of synapses (or contact area) between neurons in both directions. In Table 1, we list the network properties for both directed and symmetrized versions of the different connectomes.

To investigate covariances in neuronal activity (Fig. 1h and Supplementary Fig. 1), we study a previously published dataset of two-photon calcium recordings of large neuronal populations in the visual cortex of awake mice³². To account for variations in fluorescence across neurons, we binarize the preprocessed activity traces to reflect, at each moment in time, whether a neuron is above or below its average activity. For further details on experimental setup and data processing, see ref. 32.

Simulating the activity-independent model

In the activity-independent model (Fig. 2), we begin with n neurons, $S = \sum_{ij} A_{ij}$ units of connection strength (for an average connection strength of $\bar{s} = \frac{S}{n(n-1)}$) and a probability p of preferential growth. The network then evolves as described in the main text. Throughout the paper, we simulate networks with $n = 10^3$ neurons, for a total of $n(n-1) \approx 10^6$ possible directed connections, and (unless otherwise specified) an average connection strength of $\bar{s} = 1$. We initialize each network by distributing the strength S randomly among the possible connections. When computing distributions (as in Fig. 2b) or quantities of interest (as in Fig. 2c,d), we take 100 network samples; between each sample, we prune $n(n-1)$ different connections, such that each directed pair of neurons is pruned once on average. We also prune each connection an average of 50 times for burn-in before taking any samples. In practice, to speed up calculations, one can run multiple network updates in parallel; here we prune 1% of the possible connections at a time, and redistribute all of the pruned strength simultaneously. We confirm that this parallelization does not affect the accuracy of the simulations (Fig. 2b). In Figs. 3 and 4, to compare against the activity-dependent model, we study symmetric (as opposed to directed) networks. In practice, the only difference for the simulations is that there are $n(n-1)/2$ possible connections (neuron pairs) rather than $n(n-1)$ directed pairs, such that the total synaptic strength is given by $S = \frac{1}{2} \sum_{ij} A_{ij}$. Code for simulating the activity-independent model has been made openly available ('Code Availability').

Deriving the distribution of connection strengths

As a network evolves according to the activity-independent dynamics, the connection strengths converge to the steady-state distribution $P(s)$ defined in equation (1). To derive $P(s)$, we begin by writing down the master equation describing the change $P_{t+1}(s) - P_t(s)$ in the connection strength distribution from one time step of the dynamics t to the

next. During each network update, a random connection is pruned, yielding an average decrease in probability of $-\frac{1}{n(n-1)}P_t(s)$. On average, the pruning removes $\bar{s}$ units of connection strength. For each unit of pruned strength, we perform preferential growth with probability p , selecting each connection of strength s with probability s/S . On average, this preferential growth yields an increase in probability of $\bar{s}p\frac{s-1}{S}P_t(s-1) = \frac{1}{n(n-1)}p(s-1)P_t(s-1)$ and a decrease of $-\bar{s}p\frac{s}{S}P_t(s) = -\frac{1}{n(n-1)}psP_t(s)$. Finally, for each unit of pruned strength, we perform random growth with probability $1-p$, yielding (on average) an increase in probability of $\frac{1}{n(n-1)}\bar{s}(1-p)P_t(s-1)$ and a decrease of $-\frac{1}{n(n-1)}\bar{s}(1-p)P_t(s)$.

Combining the above contributions, we arrive at the following master equation,

$$P_{t+1}(s) - P_t(s) = \frac{1}{n(n-1)} [-P_t(s) + p((s-1)P_t(s-1) - sP_t(s)) + \bar{s}(1-p)(P_t(s-1) - P_t(s))] \quad (3)$$

As the network evolves, the connection strengths approach a steady-state distribution $P_{t+1}(s) = P_t(s) = P(s)$, which transforms equation (3) into the following recursive relation:

$$P(s) = \frac{p(s-1) + \bar{s}(1-p)}{1 + ps + \bar{s}(1-p)} P(s-1). \quad (4)$$

Solving for $P(s)$, we arrive at the distribution in equation (1), which we note is normalized throughout the paper to run over positive connection strengths $s > 0$ (that is, excluding connections of strength zero). In the limit of strong connections $s \gg \bar{s}/p$, we have the approximations $\Gamma\left[s + \bar{s}\left(\frac{1}{p} - 1\right)\right] \approx \Gamma(s)s^{\bar{s}(\frac{1}{p}-1)}$ and $\Gamma\left[s + \bar{s}\left(\frac{1}{p} - 1\right) + 1 + \frac{1}{p}\right] \approx \Gamma(s)s^{\bar{s}(\frac{1}{p}-1)+1+\frac{1}{p}}$. Thus, the distribution $P(s)$ scales asymptotically as $s^{-\gamma}$, where $\gamma = 1 + \frac{1}{p}$. The analytic steady-state distribution in equation (1) and the power-law tail are confirmed numerically in Fig. 2b. To compute the preferential probability p that best describes a given connectome (as in Fig. 2e-i), we run gradient descent (using the analytic distribution in equation (1)) on the root-mean-square error of the log probabilities.

Covariances in neuronal activity

To study activity-dependent plasticity, we implement model neurons that, between each network update, reach steady-state activities x_i defined by the nonlinear equation $x = \tanh[\beta(Ax + b)]$, where β is the interaction strength, b_i are the biases (here assumed to be zero for simplicity) and $\tanh(\cdot)$ is applied component-wise. Additionally, as is common, we implicitly normalize the connections A_{ij} by $n\bar{s}$, such that the total interaction strength per neuron approaches one for large networks. This system is equivalent to a mean-field Ising model with inverse temperature β , interactions A_{ij} and external fields $b_i = 0$. Additionally, we can map the model to an equivalent system $\tilde{x} = \sigma[\beta(\tilde{A}\tilde{x} + \tilde{b})]$ with states $\tilde{x}_i = (x_i + 1)/2 \in [0, 1]$ that represent firing rates, where $\sigma(\cdot)$ is the logistic function, and the new parameters are given by $\tilde{A}_{ij} = 4A_{ij}$ and $\tilde{b}_i = 2(b_i - \sum_j A_{ij})$.

For symmetric connections $A = A^T$, one can use the fluctuation-dissipation theorem to compute the covariances in activity,

$$C_{ij} = \frac{dx_i}{db_j} = \beta \operatorname{sech}\left[\beta\left(\sum_k A_{ik}x_k + b_i\right)\right]^2 \left(\sum_k A_{ik} \frac{dx_k}{db_j} + \delta_{ij}\right) = \beta(1 - x_i^2) \left(\sum_k A_{ik}C_{kj} + \delta_{ij}\right) \quad (5)$$

Given a stable solution x for the network activity, one can solve for the covariances $C = \beta(I - \beta DA)^{-1}D$, where $D_{ij} = (1 - x_i^2)\delta_{ij}$ and I is the identity.

The above expression for the covariances holds for arbitrary interaction strengths β and biases b . Yet for zero biases $b = 0$, things

Table 1 | Network properties of neuronal connectomes

	<i>Drosophila</i> central brain	<i>Drosophila</i> optic medulla	<i>C. elegans</i> connectome	<i>Platynereis</i> sensory motor	Mouse
<i>n</i>	21,739	1,781	279	79	1,123
<i>S</i>	1.43×10^7	3.35×10^4	6.82×10^3	1.09×10^3	$9.15 \times 10^4 \mu\text{m}^2$
$\bar{s}$	0.030	0.011	0.088	0.177	$0.073 \mu\text{m}^2$
No. of connections	$3.55 (2.90) \times 10^6$	$9.63 (8.91) \times 10^3$	$2.99 (2.29) \times 10^3$	300 (270)	$4.00 (4.00) \times 10^4$
Heterogeneity	0.616 (0.639)	0.626 (0.637)	0.433 (0.463)	0.556 (0.562)	0.451 (0.451)
Density	0.008 (0.012)	0.003 (0.006)	0.039 (0.059)	0.049 (0.088)	0.032 (0.064)
Clustering coefficient	(0.207)	(0.069)	(0.214)	(0.321)	(0.282)

For each connectome, we list the number of neurons *n*, total number of synapses (or contact area for the mouse retina) *S*, average connection strength $\bar{s}$, number of non-zero connections, connection heterogeneity, connection density and clustering coefficient. The values outside (inside) parentheses correspond to directed (symmetrized) versions of the connectomes.

simplify even further: for sufficiently weak interactions $\beta < \rho(A)$, where $\rho(\cdot)$ is the spectral radius (or the magnitude of the largest eigenvalue), the unique solution to the self-consistent equation is $x = 0$. In this case, we have $D = I$, yielding the weak-interaction expansion

$$C = \beta(I - \beta A)^{-1} = \beta \sum_{n=0}^{\infty} (\beta A)^n = \beta(I + \beta A + \beta^2 A^2 + \dots). \quad (6)$$

As the interaction strength β approaches $\rho(A)$, the solution $x = 0$ becomes unstable, and the covariances in equation (6) diverge;⁴² in this sense, $\rho(A)$ defines the critical interaction strength. For example, if the connection strength *S* is distributed randomly, then one can show that $\rho(A) = 1$ in the limit $n \rightarrow \infty$ (Supplementary Information section 3). Thus, for random connectivity, the weak-interaction expansion holds for all $\beta < 1$.

Simulating the activity-dependent model

In the activity-dependent model, the network evolves precisely as in the activity-independent model (with symmetric connections), except for one key difference: during preferential growth, rather than selecting a connection with probability proportional to its strength A_{ij} , we select a pair of neurons with probability proportional to their covariance C_{ij} . We refer to this new rule as Hebbian growth or Hebbian plasticity. Before each iteration of the dynamics, we must compute the covariances *C*, as described above. This, in turn, requires computing a solution *x* for the average activities of the neurons, which can be accomplished efficiently by initializing $x = -1$ and iterating the nonlinear equation $x \leftarrow \tanh(\beta Ax)$ until convergence⁴². In simulations, we consider networks with $n = 10^3$ neurons, for a total of $n(n-1)/2 \approx 5 \times 10^5$ possible undirected connections and (unless otherwise specified) an average connection strength $\bar{s} = 1$. As in the activity-independent model, for each calculation we record 100 samples, between which each possible connection is pruned once on average, and before which each possible connection is pruned 50 times for burn-in. In the phase diagrams in Fig. 3c–f, to improve sampling for $p \geq 0.7$, we average over ten simulations for each data point. Code for simulating the activity-dependent model has been made openly available (‘Code Availability’).

Citation diversity

Recent work in several fields of science^{55–59}, and physics in particular⁶⁰, has identified citation bias negatively impacting women and other minorities. Here we sought to proactively consider choosing references that reflect the diversity of the field in thought, form of contribution, gender and other factors. Excluding (including) self-citations to the current authors, our references contain 43% (41%) women lead authors and 36% (39%) women senior authors.

Data availability

The neuronal data analysed in this paper are openly available at github.com/ChrisWlynn/Heavy_tailed_connectivity.

Code availability

The code used to perform the analyses in this paper is openly available at github.com/ChrisWlynn/Heavy_tailed_connectivity.

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Author contributions

C.W.L. and S.E.P. conceived the project. C.W.L. designed the models, and C.W.L. and C.M.H. performed the analysis with input from S.E.P. C.W.L. wrote the paper and Supplementary Information, and C.M.H. and S.E.P. edited the paper and Supplementary Information.

Competing interests

The authors declare no competing interests.

Additional information

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